
Semiconductor Strategic Intelligence 2026–2036

Institutional Investment Committee Underwriting Document

This is an underwriting document, not a prediction report. For every company, sector, and thesis it separates business quality, investment attractiveness at today's valuation, and portfolio role. A superior business is never assumed to be a superior investment.

Market data as of mid-June 2026 and point-in-time. Decision-support analysis, not investment advice. Figures require independent verification before any allocation. Return scenarios are underwriting judgments, not forecasts or recommendations.

1 Executive Summary

The reason to spend thirty minutes on this document is that it answers, for every name and sector, the three questions an allocator actually has to resolve — is the business durable, is the price attractive, and how should the position be sized — and ties each answer to traceable evidence and an explicit exit condition. It does not recommend owning “AI.” It identifies which exposures are durable rent, which are a cycle at peak margins, which are a bet on a contested margin, and which are priced for outcomes that must go right.

Five investment conclusions

1. Durable rent is narrow and concentrated in physical and insulated-coordination chokepoints, and the return-on-capital data confirms it. TSMC sustains a return on invested capital of roughly 39–49% (2026, financial databases, T2, Medium, Estimate) on a ~50.5% operating margin (Q1 2026, company filing, T1, High, Fact); ASML and KLA sustain similarly high, stable returns. These businesses compound. Conclusion confidence: High.

2. These same businesses are high quality but, at today’s prices, offer durable rather than exceptional expected returns. Quality is largely in the price. The implication is to own them for compounding, not for a re-rating. Conclusion confidence: Medium-High.

3. Memory earnings are at a cyclical peak; the low headline multiple is a peak-earnings artifact, not cheapness. SK Hynix reported a 72% operating margin in Q1 2026 against –67% in Q1 2023 (TrendForce / company, T1–T2, High, Fact) — a 139-point swing in three years is the signature of scarcity rent. A ~7x forward multiple on a 72% peak margin is the classic cyclical trap. Conclusion confidence: Medium-High.

4. NVIDIA is contested coordination rent. Extraordinary current economics (ROIC near 150% versus a ~17% cost of capital; Jan 2026, financial database, T2, Medium, Estimate), a genuine software-and-system platform, and a ~70% gross margin (company filing, T1, High, Fact) that its largest customers are funding alternatives to erode. The investment is a position on margin durability, not a value call. Conclusion confidence: Medium.

5. The principal decade risk is a multi-year monetization shortfall, not a crash. AI adoption continues, revenue grows, returns on the installed base disappoint, and multiples compress for years without an acute trigger. The acute risk — a peripheral financing event — is real but concentrated against a well-capitalized hyperscaler core. Conclusion confidence: Medium.

What the market misunderstands, and what is priced

Consensus has moved from “AI chips are overvalued” to “the de-rating is the entry point.” The error is treating a low multiple on peak or anomalous earnings as a margin of safety. The de-rated names are partly cheap on multiple and expensive on normalized earnings; the durable compounders are fairly-to-fully valued. Cheapness and quality sit in different places, and neither coincides with the consensus long. Quality is priced in the compounders; the cyclical names price a slowdown in their multiple but not the magnitude of a margin reversion in their earnings; the contested platforms price a soft landing for margins.

Highest-conviction idea, biggest risk, portfolio implication

Highest conviction. Own the insulated compounders (TSMC, ASML, KLA, Synopsys, Cadence) as the portfolio’s durable core — not because they are cheap, but because their rent is the only rent in the sector the evidence shows compounding across the cycle. Size for valuation, and cap TSMC for the concentration tail.

Biggest risk. The monetization grind, transmitted through return-on-capital and credit indicators that lead chip earnings, against which the de-rated multiples offer less protection than they appear to.

Portfolio implication. A barbell: a durable-compounder core owned for low-variance compounding, a tactical cyclical sleeve (memory, energy bottlenecks) rented against the calendar, sized contested-platform bets (NVIDIA, Broadcom), small options (Intel, AMD), and an explicit avoid list. Sizing runs inversely to contestability and tail exposure.

2 The Semiconductor Rent Map

Durable rent comes from three sources with different decay rates, and the source predicts the decay. The single most useful diagnostic is the behavior of return on invested capital: stable high ROIC indicates durable rent; violently swinging ROIC indicates a cycle.

Physical rent — automatically insulated (ASML, TSMC, KLA)

Rooted in optics, atoms, and path-dependent yield-learning. The barrier is manufacturing complexity, cumulative yield knowledge that cannot be purchased, equipment bottlenecks, and replacement difficulty measured in years. No buyer coalition or standards body can dissolve it. ROIC is high and stable across the cycle (TSMC ~39–49%; KLA above 40% with a multi-year average near 63%; 2026, financial databases, T2, Medium, Estimate). This is where decade-length excess returns actually accumulate.

Insulated coordination rent — (Synopsys, Cadence)

Rooted not in a physical asset but in control of a coordination point: design-tool flows certified into every foundry process, with switching costs that approach the prohibitive. The complexity monopoly compounds as designs grow more complex and as systems companies become chip designers. The binding risk for these names is valuation, not moat — a critical distinction developed in Section 4.

Contested coordination rent — (NVIDIA, Broadcom)

Genuine platform strength (software ecosystem, system integration, customer relationships) that compounds with installed base but is the explicit target of buyer-funded alternatives — open software, open interconnect, and custom silicon. The rent is real and the timing of erosion uncertain, which is precisely why these positions are sized as bets on margin durability rather than underwritten as permanent. Customer alternatives are advancing; margin durability, not share, is the variable in question.

Scarcity rent — temporary by construction (HBM memory)

Rooted in a transient supply–demand imbalance. Insulated in the near term by a three-firm structure, stacking-yield difficulty, and multi-quarter qualification cycles — but depleting as announced capacity arrives around 2028, and reverting on the historical pattern. Operating margins above ~65% are without precedent in memory and do not persist. High-quality while it lasts; not a perpetuity.

Why this map may be wrong. If an insulated position’s ROIC collapses without a cyclical cause — a credible second leading-edge source taking share profitably, or a design-automation alternative breaking certification — the durability claim fails. If a scarcity position sustains peak ROIC through a full capacity cycle, memory should be re-rated toward a franchise rather than a cycle.

3 AI Infrastructure Economics

“AI risk” must be separated into three variables that do not move together: demand, financing, and technology. Conflating them produces an unhelpful boom-or-crash binary; separating them produces a monitorable framework.

Demand risk

Hyperscaler 2026 AI capex is estimated near \$700–725B, roughly double 2025 (2026, bank/media, T2–T3, Medium, Estimate), approaching ~94% of operating cash flow on a major-bank estimate (2026, bank research, T2, Medium, Estimate) — so incremental funding comes from debt markets. Pure-play AI revenue remains a fraction of capital deployed and the capital-to-revenue gap is wide and widening (2026, bank/media, T2–T3, Low-Medium, Estimate). The build is currently justified by terminal, not interim, returns. The demand case is plausible but unproven, with a thin margin of safety.

Financing risk — separate the core from the periphery

This is the distinction the popular “risk moved from valuation to financing” framing obscures. Fragility is concentrated in the periphery: asset-backed lending against rapidly depreciating accelerators, leveraged infrastructure borrowers operating on contracted backlogs rather than cash. Fragility is low at the core: the major hyperscalers carry among the strongest balance sheets in public equities and can slow spending sharply without solvency stress. The system is barbelled — a fortress core and a fragile periphery — not uniformly fragile. A peripheral credit event is a meaningful-minority probability over the horizon; its escalation to a sector event is conditional on demand deteriorating in parallel. Intercompany (“circular”) financing scale is poorly sourced and treated as speculation pending verification (2026, media, T3, Low, Speculation).

Technology risk

Accelerators carry a one-to-three-year economic life against five-to-six-year book depreciation (2026, academic/media, T2–T3, Medium, Judgment), which overstates reported earnings. Efficiency gains — architecture, sparsity, distillation, inference optimization — deflate compute demand per unit of output; the base case should assume hardware demand and capability partially decouple over the decade, moderating the silicon-volume trajectory even under strong demand.

Two measurement distortions move reported figures in opposite directions and must be corrected before either is trusted: captive hyperscaler silicon understates reported semiconductor profit, and intercompany financing overstates reported merchant demand.

Why this thesis may be wrong. If a demand pause is absorbed without credit-spread widening, the structures were more robust than their leverage implied. If the capital-to-revenue gap inflects and narrows, demand validates the build. The depreciation and capex magnitudes are T2–T3 estimates; if materially smaller, the earnings-quality concern weakens.

4 Company Underwriting

Each name separates three things — business quality, investment attractiveness at today’s price, and portfolio role — then gives an economic model, a valuation and five-year expected-return frame, the sizing logic, and the observation that would force an exit. Return ranges are five-year annualized underwriting scenarios built from a normalized-earnings path and a terminal-multiple assumption. They are not DCF outputs, not forecasts, and not recommendations. ROIC and some margin figures are T2 database estimates flagged for reconciliation to primary filings.

NVIDIA

Quality High	Attractiveness Fair (bimodal)	Portfolio role Sized platform bet
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Business quality

Contested coordination rent: CUDA software, rack-scale system integration, and an emerging enterprise layer. Pricing power high but contestable; competitive position dominant in share, weak in margin defensibility. ROIC near 150% versus a ~17% cost of capital (Jan 2026, financial database, T2, Medium, Estimate) — genuine excess returns, but unlikely to persist at that level as margin normalizes.

Economic model

Revenue growth high but decelerating from ~85% year-over-year (company filing, T1, High, Fact) toward 15–25% over the horizon (Judgment). Gross margin ~70%+ now (company, T1, High, Fact), normalizing toward the high-50s in the base case; note non-GAAP figures include stock-based compensation from Q1 FY2027 (company 8-K, T1, High, Fact), a reported-margin headwind. FCF conversion high; capital intensity low (fabless).

Valuation

~21x forward earnings on peak-margin, peak-cycle earnings (2026, market data, T2, Medium, Estimate) — not cheap, because the denominator is peak. On a high-50s normalized margin with a digestion year, the implied multiple is materially higher. The market is pricing continued high growth and margin durability: the soft-landing path.

Five-year expected-return framework

Bear	Base	Bull
–12% to 0%	6% to 11%	18% to 28%

Swing factor: gross-margin durability. Bull assumes the platform insulates and margin holds; base assumes partial normalization with share retained; bear assumes open software and custom silicon dissolve the lock-in and margin falls toward 50%.

Portfolio role

Own because it is the dominant AI-compute platform with real upside if lock-in holds; do not own more because the margin is contestable and the position is a bet, not a compounder. Size as a sized bet, monitored via custom-silicon attach and gross-margin trend.

Thesis breakers

Exit if gross margin breaks below the mid-50s while custom-silicon attach at the top-five buyers rises and open-framework inference reaches production scale. Reclassify upward if margin holds through a full custom-silicon ramp.

TSMC

Quality High	Attractiveness Fair (quality priced)	Portfolio role Core compounder (tail-capped)
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Business quality

Physical rent: leading-edge yield-learning plus advanced-packaging capability plus the commercial trust of not competing with customers; roughly 72% foundry share (2026, industry/media, T2, Medium, Estimate). Pricing power high. ROIC high and stable at ~39–49% (2026, financial databases, T2, Medium, Estimate).

Economic model

2025 revenue \$122.4B, operating income \$62.2B, net income \$55.1B (FY2025, company/filings, T1–T2, High, Fact). Gross margin ~59.6%, operating margin ~50.5% (Q1 2026, company filing, T1, High, Fact). Revenue growth mid-teens-plus on AI/HPC, normalizing. FCF pressured by Arizona and overseas capex; capital intensity high.

Valuation

~35x trailing — a compounder multiple (2026, market data, T2, Medium, Estimate). Value the durable stream at a premium and subtract an explicit geopolitical-tail discount rather than blending it into the multiple. The market prices durable leadership and AI demand; the strait tail is priced near zero.

Five-year expected-return framework

Bear	Base	Bull
Tail*	10% to 14%	18% to 22%

Swing factor: AI-demand persistence and the size of the geopolitical discount the investor demands. *The bear is bimodal: an ordinary digestion bear is roughly flat-to-low-single-digit; the rupture tail is a separate, un-modelable loss that must be sized, not averaged into the base.

Portfolio role

Own because it is the only manufacturer the evidence shows compounding across the cycle, and the integration point of the AI hardware stack. Do not own more because quality is priced and the concentration tail is unhedgeable. Cap the position size accordingly.

Thesis breakers

Exit or trim on a leading-edge yield stumble, a credible profitable second source reaching scale, or a material rise in strait risk.

ASML

Quality High	Attractiveness Fair	Portfolio role Core compounder
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Business quality

Physical monopoly: EUV lithography rests on optical and light-source physics and a supplier ecosystem no competitor is attempting to replicate. Pricing power high; ROIC high (2026, financial databases, T2, Medium, Estimate). Replacement difficulty is the highest in the industry.

Economic model

Gross margin ~52.8% (2026, market data, T2, Medium, Estimate); recent revenue growth ~31% and earnings growth ~46% (2026, market data, T2, Medium, Estimate). FCF lumpy on shipment timing; capital intensity moderate.

Valuation

~37x P/E (2026, market data, T2, Medium, Estimate). The terminal multiple is supported by monopoly economics, discounted for restricted-market revenue loss.

Five-year expected-return framework

Bear	Base	Bull
3% to 6%	10% to 13%	16% to 20%

Swing factor: restricted-market revenue retention and High-NA adoption timing. The market prices monopoly durability and steady EUV-intensity growth.

Portfolio role

Own as the purest physical monopoly in the book and a core compounder. Do not own more because the China terminal-value question and High-NA timing cap the upside.

Thesis breakers

Exit on a credible alternative EUV light source reaching volume (a next-decade risk) or a structural contraction in leading-edge fab construction.

Broadcom

Quality Medium-High	Attractiveness Expensive	Portfolio role Smaller core / tactical
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Business quality

Mixed coordination rent: a durable networking-silicon franchise plus a more-contestable custom-silicon-enablement business plus a software cash engine. Pricing power high, but the custom seam is contestable because hyperscalers can in-source the IP layer over time.

Economic model

Margins high; gross margin guided toward ~77% (2026, company/media, T2–T3, Medium, Estimate); aggressive capital return (\$7.8B repurchased with \$10B authorized; 2026, company/media, T2–T3, Medium, Estimate). Capital intensity low; ROIC high but partly acquisition-driven.

Valuation

The richest AI multiple in the group (~mid-30s forward; 2026, market data, T2–T3, Medium, Estimate), reflecting an infrastructure-platform re-rating.

Five-year expected-return framework

Bear	Base	Bull
–8% to 0%	6% to 10%	14% to 18%

Swing factor: whether the market keeps it as a platform or reclassifies it toward a cyclical supplier. The market prices durable platform status.

Portfolio role

Own smaller, as a tactical position, because the dominant risk is multiple compression rather than business deterioration. Do not own more at the current multiple.

Thesis breakers

Exit on hyperscaler in-sourcing of the enablement IP layer, custom-program cancellations, or a sustained gross-margin guide-down.

SK Hynix

Quality Medium (cyclical)	Attractiveness Trap-priced (peak earnings)	Portfolio role Tactical, dated exit
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Business quality

Scarcity rent: HBM qualification barrier and a three-firm oligopoly, re-won each generation. Pricing power high at the peak, low at the trough. ROIC durability low (cyclical).

Economic model

Q1 2026 operating margin 72%, a record exceeding TSMC; revenue KRW 52.58T, +198% year-over-year; cash KRW 54.3T (Q1 2026, company/TrendForce, T1–T2, High, Fact). Context: –67% operating margin in Q1 2023. Capital intensity high.

Valuation

~7x forward (2026, market data, T2–T3, Medium, Estimate) on peak-cycle earnings — the textbook cyclical value-trap configuration, not cheapness. The market prices a slowdown in the multiple but not the magnitude of a margin reversion.

Five-year expected-return framework

Bear	Base	Bull
–15% to –5%	6% to 11%	16% to 22%

Swing factor: cycle length versus the 2028 capacity additions.

Portfolio role

Own as a tactical position to rent the up-cycle, with a pre-set exit. Do not own more, and do not hold through the cycle, because the rent depletes and operating leverage works in reverse.

Thesis breakers

Exit on 2027 HBM contract pricing rolling over, full third-source qualification restoring price competition, or the operating margin printing more than ten points off peak.

Micron

Quality Medium (cyclical) + sovereign overlay	Attractiveness Trap-priced	Portfolio role Tactical, dated exit
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Business quality

Scarcity rent (HBM #2) plus a durable relative advantage — US-sovereign manufacturing with no restricted-market fabs, structurally valuable in a fragmenting world. Pricing power cyclical.

Economic model

Operating margin ~67.6% at cyclical peak (Q1 2026, financial database, T2, Medium, Estimate); capital intensity high.

Valuation

~11x forward (2026, market data, T2–T3, Medium, Estimate) on peak earnings.

Five-year expected-return framework

Bear	Base	Bull
–12% to –2%	7% to 12%	16% to 22%

Swing factor: cycle length plus the durability of the sovereign premium.

Portfolio role

Own as a tactical position with a dated exit; the sovereign overlay justifies a relative premium to SK Hynix in a fragmenting world. Do not own more, given reversion risk.

Thesis breakers

Same as SK Hynix; plus erosion of the sovereign premium.

KLA

Quality High	Attractiveness Fair-to-expensive	Portfolio role Core compounder (lowest variance)
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Business quality

Physical rent: process-control and metrology near-monopoly, compounding as devices go three-dimensional and inspection difficulty rises super-linearly. A recurring services floor (52+ consecutive quarters of services growth; company, T1, High, Fact) smooths the cycle. ROIC high and stable (above 40%, multi-year average ~63%; 2026, financial databases, T2, Medium, Estimate).

Economic model

Q2 FY2026 revenue \$3.30B, net income \$1.15B (company 8-K, T1, High, Fact); non-GAAP gross margin ~62.6%, operating margin ~42.8% (FY2026, company/databases, T1–T2, High, Fact/Estimate); trailing-twelve-month capital returns ~\$3.0B (company, T1, High, Fact). Capital intensity moderate.

Valuation

~45x forward (2026, market data, T2, Medium, Estimate) — a premium the ROIC and services floor partly justify.

Five-year expected-return framework

Bear	Base	Bull
5% to 7%	10% to 13%	15% to 18%

Swing factor: three-dimensional inspection intensity sustaining the equipment mix. The market prices durable process-control leadership.

Portfolio role

Own as the lowest-variance durable-rent asset in the book. Do not own more only because the entry multiple is full.

Thesis breakers

Exit on device-structure simplification (contrary to the roadmap), a metrology share loss, or a deep equipment-capex air pocket unbuffered by services.

Synopsys

Quality High (near-term drag)	Attractiveness Expensive (leverage overhang)	Portfolio role Core, size-aware
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Business quality

Insulated coordination rent — certification lock-in. Pricing power high. ROIC high structurally, depressed near-term by acquisition integration.

Economic model

Gross margin ~80% (2026, market data, T2, Medium, Estimate); reported net margin compressed to ~13.8% as of January 2026 on acquisition and debt drag (2026, financial database, T2, Medium, Estimate); ~\$10.0B long-term debt against ~\$2.5B cash post-acquisition (Apr 2026, company, T1, High, Fact); Design IP segment margin fell to ~24% from ~31% (2026, market data, T2, Medium, Estimate). Capital intensity low.

Valuation

Software multiples; the binding risk is valuation and leverage, not the moat.

Five-year expected-return framework

Bear	Base	Bull
–2% to 4%	6% to 10%	14% to 17%

Swing factor: whether the multiple holds while leverage is worked down and the acquisition integrates. The market prices a durable franchise plus a clean integration.

Portfolio role

Own as part of the insulated-coordination core, but size-aware: this is the clearest case in the book of a high-quality business that can deliver a mediocre return from a high entry point. Do not own more until the leverage and integration resolve.

Thesis breakers

Exit on a credible design-automation alternative breaking certification, integration or leverage execution failure, or restricted-market revenue runoff exceeding expectations.

Cadence

Quality High	Attractiveness Expensive	Portfolio role Core, size-aware
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Business quality

Insulated coordination rent — certification lock-in; the cleaner-margin of the two design-automation names. Benefits directly from rising design complexity and AI-driven design spend. Pricing power high; ROIC high.

Economic model

Q1 2026 revenue \$1.474B, +~19% year-over-year; non-GAAP operating margin 44.7%; record backlog \$8.0B; 2026 outlook ~17% growth (Q1 2026, company release, T1, High, Fact). Gross margin ~85% (2026, market data, T2, Medium, Estimate). Capital intensity low.

Valuation

Software multiples; valuation risk, not moat risk.

Five-year expected-return framework

Bear	Base	Bull
–2% to 4%	7% to 10%	14% to 17%

Swing factor: multiple sustainability against AI-design demand. The market prices durable growth and a sustained premium.

Portfolio role

Own as part of the insulated-coordination core, size-aware given the multiple. Do not own more at a software multiple on a spending peak.

Thesis breakers

Exit on a design-automation alternative breaking certification, AI-design demand reversing, or backlog conversion slipping.

AMD

Quality	Medium (no structural rent)	Attractiveness	Embeds share capture	Portfolio role	Optionality
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Business quality

Coordination-rent aspirant: credible design execution and a policy-driven second-source floor, but squeezed between CUDA above and custom silicon below. ROIC below the leaders, improving with mix (2026, databases, T2–T3, Low-Medium, Estimate — verify against latest filing).

Economic model

Margins improving with AI and server mix; capital intensity low. Specifics to be verified against the latest filing.

Valuation

The valuation embeds meaningful AI-share capture.

Five-year expected-return framework

Bear	Base	Bull
–10% to 0%	5% to 12%	20%+

Swing factor: share capture versus the CUDA and custom-silicon squeeze.

Portfolio role

Own as a small option on share capture rather than moat expansion. Do not own more because there is no structural rent to underwrite.

Thesis breakers

Exit if the software-ecosystem gap to CUDA fails to narrow while accelerator share at major buyers stalls.

Intel

Quality	Conditional / Low	Attractiveness	Binary	Portfolio role	Optionality / turnaround
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Business quality

Conditional rent: a product business plus a sub-scale, government-supported foundry attempting to become a second leading-edge source. ROIC: foundry loss-making; the thesis is conditional on external validation.

Economic model

Consolidated margins depressed; foundry losses with a stated later break-even path (2026, databases/media, T2–T3, Low, Estimate — verify). Capital intensity high.

Valuation

The valuation prices a turnaround probability, not a turnaround.

Five-year expected-return framework

Bear	Base	Bull
–30%+	wide and uncertain	25%+

Swing factor: the 14A external-customer decision (H2 2026–H1 2027).

Portfolio role

Own only as a small, binary-sized option. It is not a position. Do not own more than the binary justifies.

Thesis breakers

Exit on the 14A window passing without a significant external anchor, or on an 18A / 18A-P yield and ramp shortfall.

These are underwriting scenarios, not forecasts, targets, or investment recommendations. The ranges express asymmetry under stated assumptions; entry price, position size, and risk budget determine realized outcomes.

5 China and the Commodity-Tier Threat

The relevant question is not whether China reaches the manufacturing frontier but whether it can impair Western returns without doing so. It can. Pricing power dies from subsidized oversupply at adequate quality, not only from technological defeat. Frontier competition and commodity disruption must be underwritten separately.

Frontier competition — blocked this decade

China remains blocked at EUV lithography, sub-3nm yield, and leading-edge design automation, protected by physics, tacit knowledge, and export controls that are real but a wasting asset. The specific timelines for an alternative EUV source are speculation with wide error (2026, media, T3, Low, Speculation). Western frontier rent is, for now, doubly protected.

Commodity disruption — the underpriced channel

Parity is not required for economic damage. Subsidized capacity in mature-node logic, analog, microcontrollers, and power semiconductors — the last requiring no EUV — floods global markets and compresses Western margins on an engineering timetable largely independent of the AI cycle (2026, industry/media, T2–T3, Medium, Estimate). China is also entering advanced packaging and the low end of memory from below, and domestic AI silicon adequate for national needs, together with state subsidy and procurement preference, removes Chinese demand from Western results and sustains overcapacity. The pattern mirrors the rent map exactly: China cannot breach the insulated positions but methodically attacks the contestable and scarcity tiers from below. Favor the insulated frontier; underweight unsubsidized commodity-tier exposure; treat sovereign-backed capacity as a fragmentation hedge.

Why this thesis may be wrong. If China reaches frontier parity this decade, the frontier names lose their double protection and commoditization spreads upward. If commodity flooding fails to materialize — quality gaps persist, tariffs hold — the margin-assault thesis is overstated and Western mid-tier exposure is cheaper than it looks.

6 The Energy Constraint

Energy enters the thesis in two roles that a single “AI power” framing obscures. As a constraint, power availability gates compute deployment and caps the silicon volume that elevated valuations require; a credible chip-demand forecast is a power forecast first. As an opportunity, the constraint creates rent — but only where lead time is the binding factor.

Durable bottlenecks — lead time is the moat

Grid interconnection rights and queue position, large power transformers, high-efficiency turbines, and new nuclear and small-modular generation carry pricing power on the same logic as physical silicon rent: concentrated supply and multi-year lead times that capital cannot quickly relieve (2026, utility/industry, T1–T2, Medium, Estimate). When power binds, performance-per-watt becomes the dominant purchase criterion, advantaging efficiency-optimized and custom silicon and the leading-edge node.

Commodity exposure — no durable rent

Generic electrical equipment, data-center construction, and engineering-procurement suppliers are cyclical, competitive, and thin-margin regardless of the AI association. Liquid cooling is a temporary bottleneck — essential now, many entrants, falling barriers. Own the durable bottlenecks as a parallel physical-rent pool; avoid the commodity layer.

Why this thesis may be wrong. A faster-than-expected unlock of generation and grid capacity would relieve the constraint, lifting silicon volume and compressing energy-bottleneck rent. The durable-rent claim assumes lead times stay long; modular and prefabricated approaches could shorten them.

7 Scenario Framework

Six scenarios span the plausible range with interacting, not independent, variables. Probabilities are economically-justified ranges, not point forecasts. The multi-year monetization grind is treated as the central bear case.

Scenario	Type	Probability range
AI monetization success	Bull	~15–20%
Steady build	Base	~24–28%
Monetization disappointment	Bear (grind)	~17–21%
Financing-led correction	Bear (acute)	~13–17%
China commoditization	Bear (structural)	~12–16%
Geopolitical disruption	Tail	~5–9%

AI monetization success (bull). Trigger: enterprise and agentic revenue scales before financing is tested; energy keeps pace. Mechanism: returns on the installed base validate the capex and close the capital-to-revenue gap. Winners: the full rent map, with the de-rated names re-rating most. Losers: the unexposed. Implication: the repriced compounders and contested platforms are the asymmetric beneficiaries.

Steady build (base). Trigger: uneven but adequate monetization; disciplined financing. Mechanism: returns are acceptable, multiples hold, compounders compound. Winners: insulated compounders; disciplined memory traders. Losers: the over-levered periphery; perpetuity multiples on contested rent. Implication: favor durability over beta. Most of the decade probably resembles this.

Monetization disappointment (central bear). Trigger: adoption continues and revenue grows, but monetization lags capex and return on capital disappoints. Mechanism: no crash — a multi-year de-rating as the market slowly marks down terminal-return assumptions, compounded by the depreciation reckoning. Winners: design-automation and test (last budgets cut); cash-rich consolidators; eventually the users of cheap compute. Losers: merchant designers, high-multiple AI names, equipment, memory. Implication: the most damaging outcome for the consensus long, because it offers no exit signal; underweight names whose multiples assume a clean monetization ramp.

Financing-led correction (acute bear). Trigger: a peripheral credit event amid a widening gap. Mechanism: credit transmits to sentiment ahead of chip earnings. Winners: balance-sheet-strong manufacturers; design-automation and test. Losers: peripheral borrowers, memory (operating leverage), merchant designers. Implication: credit leads; hold the fortress core; the chip multiples are the lagging tell.

China commoditization (structural bear). Trigger: Chinese commodity-tier output floods, independent of the AI cycle. Mechanism: subsidized oversupply compresses Western commodity-tier margins. Winners: Chinese champions (largely uninvestable externally); the Western insulated frontier on a relative basis. Losers: Western mature, analog, power, and low-end-memory margins. Implication: a dispersion event — long the frontier, avoid the commodity tiers.

Geopolitical disruption (tail). Trigger: a disruption to the leading-edge manufacturing concentration, or a sharp controls escalation. Mechanism: demand and financing collapse together. Winners: sovereign-backed capacity; redundancy suppliers.

Losers: nearly all merchant exposure; concentration-exposed positions catastrophically. Implication: managed by sizing, not forecasting.

Why this framework may be wrong. The bands are judgments, not measurements; if monetization broadens rapidly the bull is understated, and if bifurcation arrives early the China and tail mass is understated. A genuinely novel rent pool or technology shift would sit outside the set.

8 Portfolio Construction

The framework is a barbell organized by rent type, with sizing inverse to contestability and tail exposure. The ranges below are a model framework for a dedicated semiconductor sleeve within a \$10B+ portfolio, not advice; an allocator scales them to mandate, benchmark, and risk budget.

Strategic allocation ranges (of a dedicated semiconductor sleeve)

Bucket	Range	Why own / why not more
Durable-compounder core (TSMC, ASML, KLA, Synopsys, Cadence)	~45–60%	The only rent the evidence shows compounding across the cycle. Not more because quality is priced and TSMC carries the concentration tail.
Cyclical tactical (memory, durable energy bottlenecks)	~10–20%	Asymmetric up-cycle return and a parallel physical-rent pool. Not more because of peak-cycle earnings and reversion risk; must be exit-disciplined.
Contested-platform bets (NVIDIA, Broadcom)	~10–20%	Dominant platforms with real upside if lock-in holds. Not more because the margin and multiple are contestable.
Options (Intel, AMD)	~0–5%	Asymmetric binary and share-capture payoffs. Not more because there is no structural rent.
Cash / flex	residual	Deployed on dislocations.

Maximum position sizes and concentration limits

No single name above ~10–12% of the sleeve. TSMC capped tighter at ~8–10% despite its quality, because the concentration tail is unhedgeable and uncorrelated with the rest of the thesis. No cyclical name (memory) above ~5–6%, given reversion risk. No option (Intel, AMD) above ~2–3%, given binary outcomes. Three concentration risks are managed explicitly: (1) the Taiwan / leading-edge geographic concentration — the largest non-diversifiable risk, managed by capping TSMC, preferring re-deployable equipment, materials, and design tools over physical fabs at the margin, and sizing to survive a prolonged adverse period without forced selling; (2) AI-demand factor concentration — most of the sleeve is one bet on AI monetization, with the durable-compounder core and design-automation/test as the partial hedges that survive the grind; (3) financing-periphery contagion — underweight leverage- and intercompany-dependent positions, with the core hyperscalers' balance sheets as the system's stabilizer.

Risk-budget logic and exposure conditions

Spend the risk budget where rent is insulated and the payoff compounds (the core); rent, do not own, where rent depletes (the cyclical sleeve); buy optionality cheaply where rent is contested or binary (bets and options). Increase exposure to the cyclical sleeve only early in a memory up-cycle with a pre-set exit; to the contested-platform bets only on evidence of margin durability; and to the core on multiple compression that is not accompanied by ROIC deterioration — a valuation gift, not a quality break. Decrease exposure on a widening capital-to-revenue gap with soft utilization (the grind confirming), peripheral credit-spread widening (the acute correction confirming), or memory operating margins printing more than ten points off peak (the cycle turning).

Why this is not simply an AI basket

The consensus AI basket conflates business quality, valuation, and portfolio role, and is implicitly one undiversified bet on AI monetization at multiples that assume a clean ramp. This construction separates the three layers, concentrates risk where rent is durable, rents the cycle rather than owning it, sizes the AI-margin bet rather than assuming it, and pre-commits to the indicators that lead chip earnings. It is built to compound in the base and bull cases and to survive the grind, the financing correction, and the commoditization shock with bounded loss.

9 Investment Committee One-Pager

Thesis. Durable semiconductor rent is narrow — physical (ASML, TSMC, KLA) and insulated-coordination (Synopsys, Cadence). Contested coordination rent (NVIDIA, partly Broadcom) is valuable but not permanent. Memory is a cycle at peak margins, cheap on multiple and expensive on earnings. The principal decade risk is a multi-year monetization grind, not a crash; the acute risk is a peripheral financing event against a fortress hyperscaler core.

Three-layer summary. Best quality: TSMC, ASML, KLA, Synopsys, Cadence. Best attractiveness: those names, but only as durable compounding (quality is priced); the de-rated chips are cheap on multiple, dangerous on earnings. Roles: core compounders (the five), tactical cyclicals (memory, energy bottlenecks), sized bets (NVIDIA, Broadcom), options (Intel, AMD), avoid (unsubsidized commodity tiers).

Positioning. Core ~45–60%, cyclical ~10–20%, bets ~10–20%, options ~0–5%; maximum single name ~10–12%, TSMC capped ~8–10% for the tail.

What changes the view. Up: monetization closes the gap; NVIDIA margin holds through a custom-silicon ramp; core multiples compress without ROIC deterioration. Down: the capital-to-revenue gap widens with soft utilization; peripheral credit spreads widen; memory margins fall ten-plus points off peak.

Caveat. ROIC and several macro figures are T2–T3 estimates pending reconciliation to filings; return ranges are underwriting scenarios, not DCF outputs or targets; at least one of the decade’s material rent pools is likely unnamed here.

A Appendix A — Evidence Ledger

Every important figure mapped to claim, source, date, tier, and confidence. No Volume I posture rests solely on a Tier 3 / Speculation row; where such an input is material, the text hedges and the Assumption Register records the dependency.

Claim	Source	Date	Tier	Conf.
TSMC 2025 rev \$122.4B / OI \$62.2B / NI \$55.1B	Company / filings	FY2025	T1–T2	High
TSMC ROIC ~39–49%	Financial databases	2026	T2	Med
TSMC gross ~59.6% / operating ~50.5%	Company filing	Q1 2026	T1	High
NVIDIA ROIC ~150% vs WACC ~17.3%	Financial database	Jan 2026	T2	Med
NVIDIA Q4 FY26 FCF ~\$34.9B; gross ~70%+	Company 8-K	FY2026	T1	High
NVIDIA non-GAAP includes SBC from Q1 FY27	Company 8-K	FY2026	T1	High
SK Hynix Q1 2026 operating margin 72%	Company / TrendForce	Q1 2026	T1–T2	High
SK Hynix Q1 2023 operating margin ~67%	TrendForce	Q1 2023	T2	High
Micron operating margin ~67.6%	Financial database	Q1 2026	T2	Med
ASML gross ~52.8%, P/E ~37x	Market data	2026	T2	Med
KLA Q2 FY26 rev \$3.30B / NI \$1.15B; nGM ~62.6%	Company 8-K	FY2026	T1	High
KLA ROIC >40% (multi-yr avg ~63%)	Financial databases	2026	T2	Med

Claim	Source	Date	Tier	Conf.
Cadence Q1 26 rev \$1.474B (+19%), nOM 44.7%, backlog \$8.0B	Company release	Q1 2026	T1	High
Synopsys ~\$10.0B LT debt vs ~\$2.5B cash; net margin ~13.8%	Company / database	2026	T1–T2	Med-High
Broadcom GM guided ~77%; \$7.8B buyback, \$10B authorized	Company / media	2026	T2–T3	Med
Hyperscaler 2026 AI capex ~\$700–725B	Bank / media	2026	T2–T3	Med
Capex ~94% of operating cash flow	Bank research	2026	T2	Med
Capital-to-revenue gap wide / widening	Bank / media	2026	T2–T3	Low-Med
GPU 1–3y life vs 5–6y book depreciation	Academic / media	2026	T2–T3	Med
Intercompany (“circular”) financing scale	Media	2026	T3	Low
China alt-EUV timelines; SMIC yields	Media	2026	T3	Low

B Appendix B — Source Hierarchy

Tier 1 (admissible for committee figures): company filings, SEC filings, earnings releases, government data, IEA, official semiconductor bodies (SIA, WSTS). **Tier 2 (estimates, must be attributed):** major-bank and consulting research, Gartner, peer-reviewed academic work, established financial-data aggregators (used here for ROIC and some margins where filings were not directly parsed; flagged for reconciliation). **Tier 3 (directional only, never the sole basis for a posture):** media, market commentary, secondary estimates. In this edition, company results and margins are T1; ROIC and some normalized margins are T2; capex and financing macro and China timelines are T2–T3; intercompany-financing scale is T3.

C Appendix C — Company Underwriting Methodology

The per-company models in Section 4 follow a fixed structure: business quality (moat, rent type, pricing power, ROIC durability), economic model (revenue drivers, margin structure, capital intensity, FCF, cyclicity), valuation (current multiple, normalized earnings, what is priced), a five-year expected-return frame (bear/base/bull with the swing factor), portfolio role, and thesis breakers. The five-year annualized return ranges are built from a normalized-earnings path (revenue growth multiplied by normalized margin) and a terminal-multiple assumption, cross-checked against the current multiple. This is underwriting, not discounted cash flow; the explicit limitation is that FCF timing, buybacks, and balance-sheet effects are approximated, and ROIC inputs are T2 estimates pending reconciliation to primary filings. The ranges express asymmetry under stated assumptions, not point forecasts, targets, or recommendations.

D Appendix D — Historical Analogues

History is used to test mechanisms, not to narrate or to draw superficial comparisons.

Episode	Mechanism tested (similarity / difference)	Investment lesson
DRAM / NAND cycles	Oligopoly discipline versus capacity-driven reversion. Similar: HBM is an oligopoly with qualification barriers and margins at a cyclical peak. Different: co-design adds a temporary, single-generation switching cost.	Memory operating margins above ~65% revert; size memory as a cycle with a dated exit, not as a franchise.

Episode	Mechanism tested (similarity / difference)	Investment lesson
Telecom / fiber 1996–2002	Real demand built ahead of monetization, debt-financed. Similar: capacity ahead of revenue; rising leverage in the deployment layer. Different: today's core hyperscalers are far better capitalized than the 1990s carriers.	Real demand does not prevent capital destruction when financing duration mismatches the monetization timeline; watch the periphery, not the core.
Intel manufacturing decline	Learning-curve leadership is losable. Similar: leading-edge yield is the moat and a single misstep compounds. Different: TSMC's no-compete-with-customers trust position differs from integrated Intel.	Manufacturing leadership compounds but is not permanent; monitor yield and the ROIC trend.
Proprietary versus open standards	Important interfaces standardize over time. Similar: proprietary interconnect and software (NVLink, CUDA) under coordinated buyer attack. Different: software and ecosystem lock-in has historically been deeper than hardware-interface lock-in.	Contested coordination rent decays on a lag; do not capitalize it as a perpetuity.
Cloud infrastructure 2006–2015	Heavy upfront capex preceding durable returns. Similar: large capital build ahead of monetization. Different: cloud monetized into proven recurring software and services economics; AI capex is heavier and faster.	Terminal-return justification is credible only when interim unit economics are visible; demand the ROIC path, not the promise.

E Appendix E — Assumption Register

Assumption	Importance	Current belief	What would invalidate it
AI demand growth persists	High	Plausible, unproven	Capital-to-revenue gap widens into 2027 with soft utilization
NVIDIA gross-margin durability	High	Normalizes to high-50s	Holds through a full custom-silicon ramp (bull) or breaks below 50% (bear)
CUDA / coordination durability	High	Eroding at the inference margin	Open-framework parity reaches production scale
Memory reverts on 2028 capacity	High	Base case (margins at peak)	Discipline and pricing hold through the additions with multi-generation switching costs
Financing: periphery vs core	High	Concentrated fragility, fortress core	A peripheral credit event contaminates sentiment broadly
TSMC geopolitical continuity	High	Assumed in base; tail un-priced	A strait disruption (un-modelable)
EDA valuation holds	Medium	Software multiples; Synopsys leverage drag	Multiple compresses faster than earnings compound
China stays sub-frontier	Medium	Blocked this decade	Alternative EUV at volume earlier than expected
Energy unlocks in time	Medium	Partly power-gated	A fast generation and grid unlock

F Appendix F — Monitoring Dashboard

Indicators that lead chip earnings, with frequency, trigger level, and investment implication. Escalation rule: review posture on two or more bear readings in the financing or returns categories, or on any memory operating-margin print more than ten points off peak.

Indicator	Freq.	Bear trigger level	Investment implication
Capital-to-revenue gap	Q	Widens into 2027 with soft utilization	Underweight high-multiple AI names; grind confirming
Return-on-capital trend	Q	ROIC inflects down across AI names	Trim contested platforms and equipment
Asset-backed accelerator spreads	M	Spreads widen; deals undersubscribed	Reduce periphery; raise quality

Indicator	Freq.	Bear trigger level	Investment implication
Peripheral-borrower credit spreads	M	Move toward distressed levels	Acute correction confirming; hold core
Hyperscaler capex / op. cash flow	Q	Sustained above 100% with negative FCF	Funding stress; de-risk deployment exposure
HBM 2027 contract pricing	Q	Rolls over	Exit memory tactical sleeve
Memory operating margin vs peak	Q	More than ten points off peak	Cycle turning; complete memory exit
Foundry second-source decision (14A)	Event	Window passes without an anchor	Close the Intel option
China commodity-tier export pricing	Q	Accelerating declines	Avoid commodity tiers; favor frontier
Transformer / interconnection lead times	Q	Extend further	Tilt to durable energy bottlenecks

Q CIO Five-Questions Quality Test

The document is built to answer, for any conclusion, the five questions a skeptical CIO would ask before allocating capital. Each is addressed identically throughout.

#	Question	How this document answers it
1	Can the CIO identify exactly why each position exists?	Every name carries a rent type, a three-layer verdict (quality, attractiveness, role), and a stated portfolio purpose with sizing logic.
2	Can the CIO identify what would invalidate each thesis?	Every name has specific, measurable thesis breakers; every chapter has an explicit “why this thesis may be wrong.”
3	Can the CIO distinguish company quality from expected return?	The three-layer separation is enforced everywhere, with the explicit finding that the highest-quality names are fairly-to-fully priced.
4	Can the CIO understand the portfolio sizing logic?	Section 8 provides allocation ranges, maximum position sizes, concentration limits, risk-budget logic, and increase / decrease conditions.
5	Can every important number be traced to evidence?	Every figure carries a source tier, confidence level, and Fact / Estimate / Judgment / Speculation label, and maps to the Evidence Ledger in Appendix A.

Residual limitations, stated plainly. ROIC and several macro inputs are T2–T3 estimates pending reconciliation to primary filings; return ranges are underwriting scenarios, not DCF outputs, targets, or recommendations; probabilities are economically-reasoned ranges, not measurements; and the framework extrapolates today’s rent pools, so at least one material future rent pool is likely unnamed. These do not change the postures but bound the confidence with which they are held.

Decision-support analysis, not investment advice. All figures are point-in-time (mid-June 2026) and require independent verification before allocation.